Oil Spill SAR Image Segmentation via Probability Distribution Modeling

Fang Chen Aihua Zhang Heiko Balzter Peng Ren
Huiyu Zhou

1 Introduction

Marine oil spills cause severe environmental damage and pose significant threats to marine ecosystems and coastal regions. Rapid and accurate detection of oil spills is therefore essential for environmental monitoring and disaster mitigation. Synthetic Aperture Radar (SAR) imagery plays a crucial role in marine oil-spill detection due to its all-weather, day-and-night imaging capability.

Oil spills alter the ocean surface by suppressing capillary and short gravity waves, resulting in dark regions in SAR images dominated by non-Bragg scattering. While many existing methods focus on oil-spill detection from either physical modeling or image processing perspectives, segmentation accuracy remains challenging due to SAR image noise, irregular spill shapes, and complex ocean conditions.

Traditional oil-spill segmentation approaches include thresholding, clustering, active contour models, and deep learning techniques. However, many of these methods rely heavily on visual features and do not explicitly consider SAR image formation mechanisms. Moreover, segmentation accuracy is often sensitive to initialization and image quality.

To address these challenges, this work proposes a novel segmentation framework that integrates SAR image formation modeling with oil-spill segmentation. By exploiting the probability distribution representation of SAR imagery, the proposed method constructs a segmentation energy functional that directly incorporates oil-spill physical characteristics, enabling accurate and robust oil-spill segmentation.

2 SAR Imaging for Marine Oil Spills

SAR images are generated by measuring microwave backscattering from the Earth's surface. The presence of oil spills modifies the scattering mechanism by damping surface roughness, resulting in lower backscattered power compared to surrounding clean water areas.

Based on the SAR imaging mechanism, the observed intensity of SAR images follows a probabilistic distribution related to the radar cross section (RCS). The paper demonstrates that SAR image intensity can be effectively modeled using an exponential probability distribution, which captures the physical backscattering behavior of oil-covered sea surfaces.

Although Weibull and Gamma distributions are commonly used to describe SAR imagery, these distributions are less suitable for oil-spill modeling. The Weibull distribution tends to misrepresent low-intensity regions, while the Gamma distribution primarily models speckle noise rather than oil-spill characteristics. Consequently, the exponential distribution is adopted as the most appropriate model for representing oil-spill SAR images.

This probabilistic representation provides a solid foundation for integrating SAR image formation into the segmentation framework.

3 Energy Functional for Oil Spill Segmentation

Oil-spill segmentation is formulated as an energy minimization problem. The image domain is divided into oil-spill and non-oil-spill regions, each characterized by distinct probability distributions.

Using the exponential distribution model, a segmentation-fitting energy term is constructed based on the negative log-likelihood of SAR image intensities. This term encourages the segmentation process to align with the statistical properties of oil-spill and background regions.

To ensure that segmentation is performed across the entire image domain while preserving local details, a Gaussian kernel function is introduced. This kernel adapts the influence of neighboring pixels and improves segmentation robustness under intensity inhomogeneity.

The resulting energy functional measures how well the segmentation aligns with the underlying SAR image distribution, providing a principled and physically meaningful segmentation criterion.

4 Level Set Formulation

To represent oil-spill boundaries accurately, the segmentation framework employs a level set method. In this formulation, oil-spill contours are represented implicitly as the zero-level set of a continuous function.

The level set formulation allows the segmentation contour to evolve smoothly and handle complex topological changes, such as merging or splitting oil-spill regions. A Heaviside function is used to distinguish oil-spill and background regions during energy computation.

By embedding the probability-based fitting energy into the level set framework, the segmentation process is guided by both statistical image properties and geometric contour evolution.

5 Regularization Terms

Due to the irregular shapes of oil spills, regularization terms are incorporated into the segmentation energy functional to ensure numerical stability and contour smoothness.

A contour regularization term penalizes excessively complex boundaries by approximating the length of the oil-spill contour. This prevents overfitting to noise and maintains realistic spill shapes.

Additionally, a distance regularization term is introduced to preserve the signed-distance property of the level set function. This regularization improves numerical stability and eliminates the need for reinitialization during contour evolution.

Together, these regularization terms ensure smooth, stable, and accurate segmentation results.

6 Overall Segmentation Energy Functional

The complete segmentation energy functional consists of three components:

- A probability-based fitting energy term derived from SAR image distribution modeling,
- A contour regularization term enforcing boundary smoothness,

- A level set regularization term ensuring numerical stability.

These components are combined into a unified energy functional, whose minimization yields the final oil-spill segmentation. Balancing parameters are used to control the relative influence of each component.

This integrated formulation allows SAR image formation and segmentation to be addressed simultaneously within a single optimization framework.

7 Energy Functional Minimization

Segmentation is achieved by minimizing the overall energy functional using an alternating optimization strategy. The minimization process consists of two steps performed iteratively.

First, the statistical parameters representing oil-spill and background distributions are updated analytically. These parameters guide the segmentation toward regions consistent with oil-spill intensity statistics.

Second, the level set function is updated using a gradient descent scheme derived from the Euler–Lagrange equation. This update drives the contour evolution toward optimal oil-spill boundaries.

The iterative alternation between parameter estimation and contour evolution ensures convergence to a stable and accurate segmentation result.

8 Methodology

The proposed methodology integrates SAR image formation theory with variational image segmentation. The framework begins by modeling SAR oil-spill imagery using an exponential probability distribution that reflects physical scattering behavior.

This distribution model is then embedded into a level set–based segmentation framework through a carefully designed energy functional. Regularization terms ensure smooth contour evolution and numerical stability.

An alternating minimization algorithm is employed to optimize both distribution parameters and segmentation contours. This synergy between physical modeling and variational optimization enables robust oil-spill segmentation across diverse SAR datasets.

9 Performance Analysis

Extensive experiments are conducted on SAR datasets obtained from ERS-1, ERS-2, and Envisat satellites. Both visual and quantitative evaluations demonstrate that the proposed method outperforms state-of-the-art segmentation techniques.

The method achieves higher segmentation accuracy and precision compared to active contour models, domain transformation approaches, and neural network–based methods such as U-Net. It also exhibits faster convergence and greater robustness to initialization variations.

Further comparisons with Weibull- and Gamma-based segmentation models confirm that the exponential distribution provides superior oil-spill characterization. Overall, the results validate the effectiveness, stability, and practical applicability of the proposed segmentation framework for real-world marine oil-spill monitoring.